



Investigation of the use of near synonyms: strong Vs. powerful

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Introduction

This research project will explore the use of the near synonyms “strong” and “powerful”.

Near-synonyms are “words which designate the same thing but emphasize different aspects of it, or words which have the same meaning, but differ in its finer shades.” (Apresjan, 1973: 175).

The definitions of the two words provided by the Oxford Learner’s Dictionary are: **strong:** having a lot of physical power; having great power; having a powerful effect on the body or mind; having a lot of power or influence; **powerful:** being able to control and influence people and events; having great power or force; very effective; having a strong effect on your mind or body; physically strong. Comparing them, it can be noted that the meanings of the two words are the same, what changes is their order: strong firstly indicates a physical force, whereas powerful firstly indicates a non-physical force. In examples 1 and 2 it can be clearly seen that these words are near synonyms because they can be switched without changing the overall meaning of the utterances.

1. “The wind got up swiftly here and it was already quite *strong* tonight.”
2. “She tensed with the need to resist the *powerful* feelings he was evoking.”

The research question of this study is: what motivates the choice between the use of “strong” and “powerful”? The choice of this research question comes from the widespread use of the word strong. In fact, it was believed interesting to explore the shades in its meaning and to compare it to its near synonyms.

It should be mentioned that this topic has been **previously explored** in Church, Gale, Hanks, & Hindle, (1991) who state that “An important criterion for differentiation seems to be that strong tends to denote an intrinsic quality whereas powerful appears to be extrinsic, referring

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more to the effect on others or on the external world.” (p. 133). These prior findings were considered during the investigation.

The use of the two words is likely to be motivated by both collocational preferences and by inter- and extra-linguistic factors. Due to the extensive number of such factors, this research focused only on two of them. The variables retained were the perceived level of difficulty (BNCweb annotation) and intrinsicness/extrinsicness (self-annotation). Two hypotheses were formulated. The first hypothesis states that strong is more frequent in texts which display a low level of difficulty, whereas powerful is more frequent in high-difficulty texts. The second hypothesis is linked to the previous literature and claims that strong is associated with the variable of intrinsicness, while powerful is associated with the variable of extrinsicness. Simply put, words which collocate with strong are hypothesised to have inherent qualities, whereas linguistic items which collocate with powerful are hypothesised to have an impact on others.

Take for example the following two sentences:

- 1) “Take *strong* and comfortable shoes –you will find cobbled streets everywhere.”
- 2) “The pressure on these sites for leisure and recreation uses is becoming increasingly *strong* in the county.”

In 1) the quality of the shoes is intrinsic, whereas in 2) the pressure has extrinsic features.

Methods

To carry out this research, the British National Corpus was used. BNC is a large corpus of contemporary British English containing 4000 text files together with their metadata (e.g. the age of the author, the gender of the author, etc.). It is PoS-tagged, lemmatized but not syntactically parsed. Data were retrieved with BNCweb, which is “a web-based client program

for searching and retrieving lexical, grammatical and textual data from the British National Corpus (BNC)”¹. Two separate collocational analyses were performed restricted to “any noun”. For the bivariate analysis, the query algorithm used was: (strong_{ADJ}|powerful_{ADJ}). Data cleaning, data annotation, and bivariate analysis were carried out in MS Excel. During the process of data cleaning, the observations for which there was no annotation were discarded and the annotation was made consistent. Regarding the variable of intrinsicness/extrinsicness, observations were randomized to get a representative sample, and 200 of them were selected and annotated. Concerning operationalization, adjectives were considered intrinsic if in the context/cotext in which they were they displayed an inherent characteristic, and they were considered extrinsic if in the context/cotext in which they were they referred to the effect on others or on the external world. The statistical analysis was performed with VassarStats and comprises a Chi-squared test of independence, the p-value, the effect size, and the standardized residuals.

Results

As far as the collocational analysis is concerned, according to the data retrieved from BNC, strong tends to collocate with nouns belonging to the semantic fields of natural forces (e.g.: wind, breeze, ...), senses (e.g.: smell, light, ...), and statistical terminology (e.g.: tendency, growth, ...), whereas powerful tends to collocate with nouns belonging to the semantic fields of tools and machines (e.g.: tool, engine, ...), politics (e.g.: politician, faction, ...), war (e.g.: ally, enemy, ...).

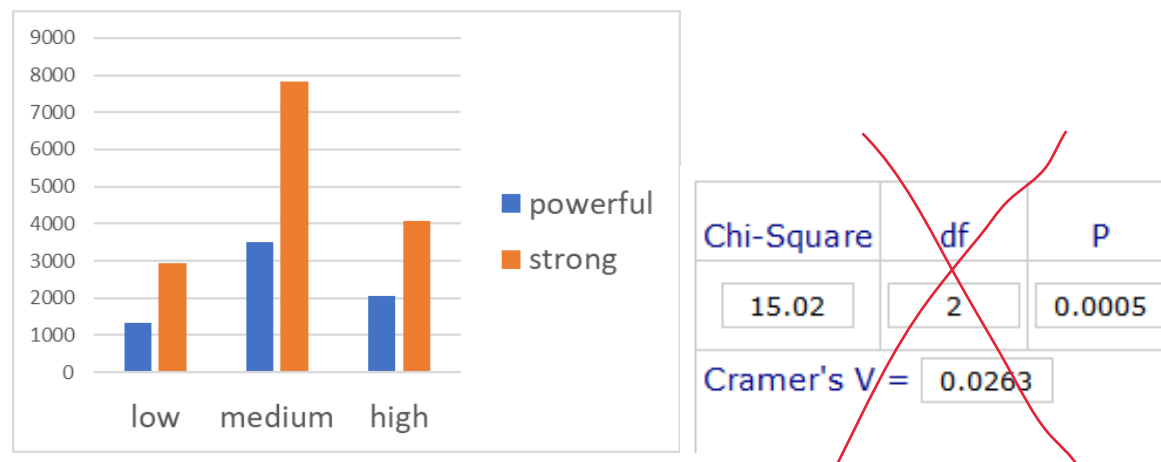
¹ <http://rdues.bcu.ac.uk/bncweb/manual/bncwebman-home.htm>

~~The bivariate analysis comprising cross-tables will now be presented for each variable.~~ As far as the perceived level of difficulty is concerned, it was found evidence of a higher frequency of strong compared to powerful for all the perceived levels of difficulty. ~~Performing the statistical analysis, a significant P-value was found ($< 5\%$).~~ However, the effect size was small (Cramer's $V = 0.0263$). ~~Finally, checking the standardized residuals, evidence was found of a positive association between powerful and low level of difficulty (< 2).~~ To summarize, the Chi-squared test of independence showed that there is a statistically significant association between the choice of the adjective and the perceived level of difficulty.



Difficulty	Adjective	
	Powerful	Strong
High	1329 (31%, -0.68)	2939 (69%, +0.46)
Medium	3506 (31%, -1.57)	7843 (69%, +1.07)
Low	2065 (34%, +2.71)	4068 (66%, -1.84)

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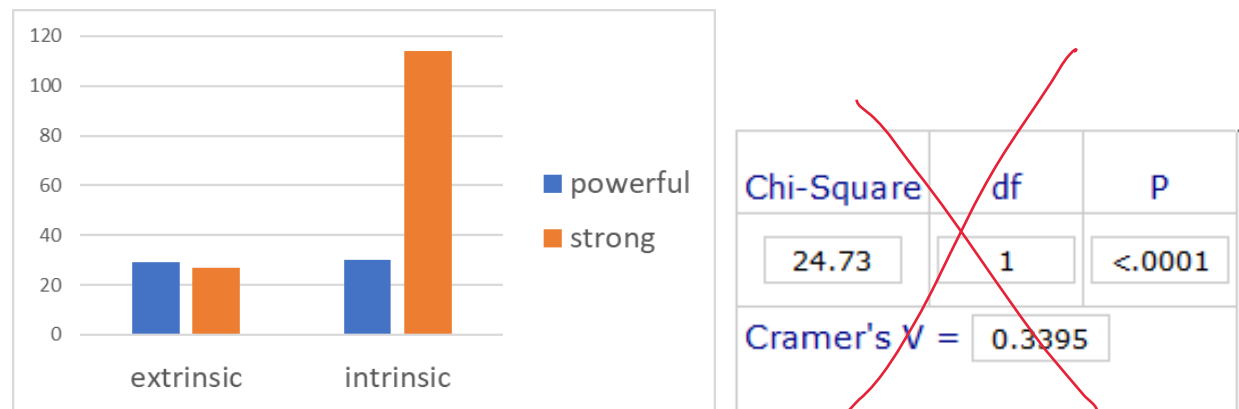
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As far as the second variable is concerned, based on the BNC data, evidence was found that strong is more often used with an intrinsic value whereas powerful is more often used with an extrinsic value. After having carried out the statistical analysis, it was found a significant P-value ($< 5\%$). However, the effect size was small (Cramer's $V = 0.3395$). Finally, checking the

standardized residuals, evidence was found that there is a positive association between powerful and extrinsic (>2), and a strong negative association between both powerful and intrinsic and strong and extrinsic (<-2). To summarize, the Chi-squared test of independence showed that there is a statistically significant association between the choice of the adjective and the variable of intrinsicness/extrinsicness.

Intrinsicness/extrinsicness	Adjective	
	Powerful	Strong
Extrinsic	29 (52%, +3.73)	27 (48%, -2.19)
Intrinsic	30 (21%, -2.12)	114 (79%, +1.24)

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Discussion

This study investigated the use of the near-synonyms strong and powerful. According to the data retrieved from BNC, evidence was found of a statistically significant association between powerful and low level of difficulty, and between strong and intrinsicness and powerful and extrinsicness. With respect to the first hypothesis, it was rejected. Indeed, evidence was found of an association between the use of powerful and low level of difficulty. Regarding the second hypothesis, the tendencies fit to half of it as evidence was found of an association

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between powerful and extrinsicness. These findings provide a clearer understanding of the meaning of the two near-synonyms and lead to a more conscious use.

Conclusion

To conclude, it is likely that the choice between “strong” and “powerful” is motivated by multiple variables working at the same time. The present study found a significant positive association between the choice of strong the low level of difficulty and intrinsicness, and the choice of powerful and extrinsicness.

Nevertheless, only two variables have been taken into consideration, one of which was self-annotated. It would be interesting to have the sample annotated according to the intrinsicness/extrinsicness variable by someone else and to calculate Cohen’s Kappa in order to evaluate the degree of agreement. Furthermore, an investigation of other variables such as the age of the user or the text genre could lead to a more detailed comparison and a broader understanding of their meanings.

References

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Data cited herein have been extracted from the British National Corpus, distributed by the University of Oxford on behalf of the BNC Consortium. All rights in the texts cited are reserved.

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Code book

• Creation of the dataset:

- Author(s) & Date [REDACTED], December 2019
- Name of research project: Investigation of the use of near-synonyms: strong Vs. powerful
- Objectives of the research project: explore differences in the meanings of strong and powerful
- Corpus Name: BNC
- Software that was used to retrieve, clean and annotate the data: BNCweb (retrieve), MS Excel (clean and annotate)
- Query: (strong_{ADJ}|powerful_{ADJ})
- Data retrieved by: [REDACTED]
- Data annotated by: BNC (perceived level of difficulty), [REDACTED]
(intrinsicness/extrinsicness)

• Information about the final dataset:

- Variable names, values and the operationalization: perceived level of difficulty (low, medium, high), intrinsicness/extrinsicness (intrinsic, extrinsic)